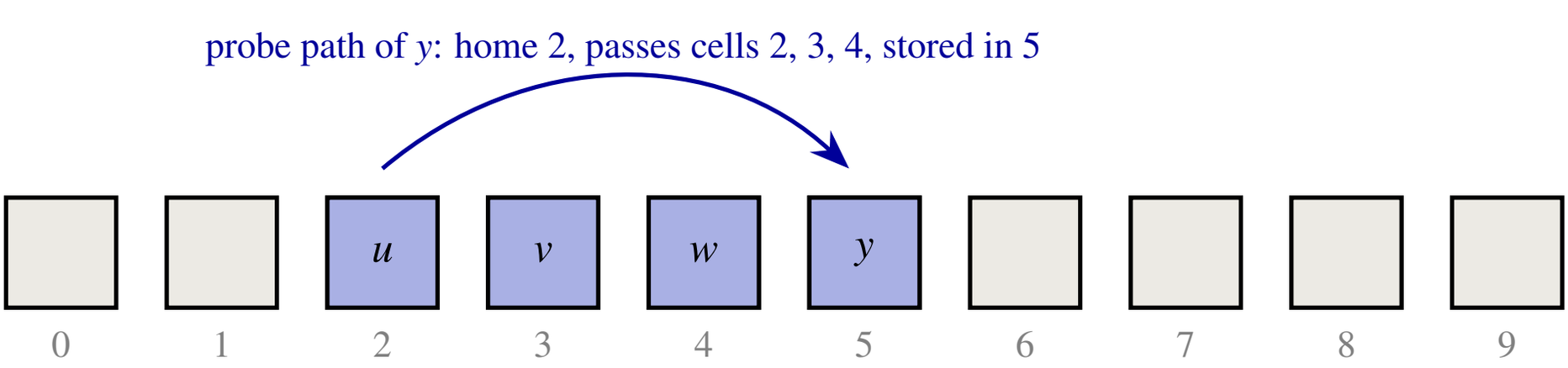
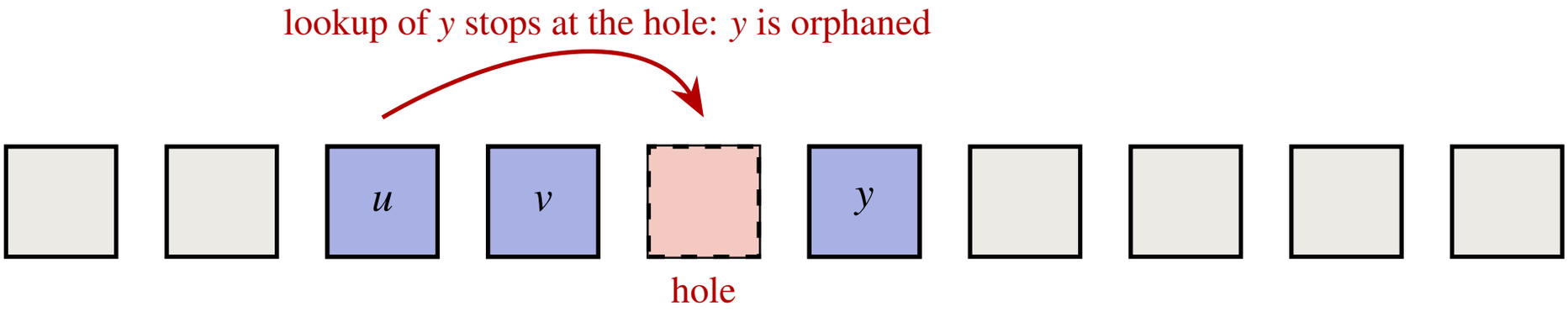


Dependency: $y \lhd x$ if the cell of x lies strictly before the cell of y on the probe path of y.
Here $y \lhd u$, $y \lhd v$, $y \lhd w$. A deletion order is safe iff every key is cleared before the keys it depends on (linear extension of $\lhd$).

Clearing w before y (not stack order):



Stack order is always safe: the newest key has no dependents, because a key's path crosses only cells that were filled before it. It is also the only order safe for every hash assignment (safe-orders theorem, part b). The number of keys orphaned by clearing one key is a queue length minus one (orphan law).